Taylor McDonald

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Education

New Jersey Institute of Technology — Newark, NJ
Bachelor of Science in Data Science — *Expected Dec. 2027*

- Relevant Coursework: Database System Design and Management (SQL), Fundamentals and Principles of Data Science, Introduction to Artificial Intelligence, Programming Language Concepts (C++), Computers, Society and Ethics

Hanyang University ERICA — Ansan, South Korea
Study Abroad Program — *Spring 2025*

Experience

Undergraduate Research Intern, Machine Learning — Jun. 2026 – Aug. 2026
SETI Institute, Carl Sagan Center — *Remote*

- Delivered a working **Python** version of an NSF-funded MATLAB sea ice forecasting codebase, matching ground truth on **RMSE** across all nine core functions, by rewriting each function and fixing three critical bugs
- Produced a September 2024 Arctic sea ice extent forecast of about 4.47 million km^2 for a UCLA and UCSB collaboration, by aggregating satellite and reanalysis data with **pandas** and **NumPy**
- Made the full pipeline reproducible for the research team, delivered as nine documented **Jupyter** notebooks with FAIR compliance files, by writing step-by-step documentation; co-authoring a manuscript

Founder and Owner — Aug. 2025 – Present
All The Paws LLC — *East Orange, NJ*

- Raised average revenue per client by 10 to 15% across about 30 households, by analyzing service history and redesigning pricing for 6 services with subscription discount tiers
- Retained 100% of clients through the September 2026 price change, by explaining every update in a plain-language client bulletin and a rewritten service agreement
- Moved client booking online by launching allthepaws.co (GitHub Pages, custom DNS) and integrating Walkies, replacing scheduling by text and phone

Projects

Melting Shapes: iOS Climate Education Game | *Swift, SwiftUI, Xcode, Python* — Jul. 2026 – Present

- Taught Arctic ice-albedo feedback in a 3,200-line iOS game for the SETI Institute, by adapting researcher Ivan Sudakow's board game into a full Swift engine with 20 shapes per color, rule validation, and a computer opponent
- Made climate vs. weather measurable for players, through score history and tiered results grounded in NASA and NSIDC sea ice data, by designing a mode where probability-weighted weather cards change scoring
- Preparing App Store launch and funding with Rose Rocchio, Director of Mobile Web Research and Accessibility, UCLA Office of Advanced Research Computing; added English and Korean localization

Marvel Rivals Hero Tier List Predictor | *Python, KMeans, Streamlit, Git* — 2026

- Ranked 40 heroes into 5 tiers, by collecting match data from a public API, engineering role-normalized metrics so heroes compared fairly across roles, and clustering with **KMeans**
- Shipped an interactive **Streamlit** dashboard to explore the tiers, managing the build with Git branches and pull requests

PawsIQ: Demand and Churn Analytics | *Python, XGBoost, scikit-learn, FastAPI, React* — In Progress

- Building demand forecasting and client churn models in **XGBoost** on real booking data from about 30 All The Paws households

Technical Skills

Languages: SQL, Python, R, Java, C++, MATLAB, Swift
Data and Analytics: pandas, NumPy, scikit-learn, XGBoost, SHAP, statistical analysis, time series forecasting
Tools: Microsoft Excel, Word, PowerPoint, Jupyter, Xcode, SwiftUI, Git, GitHub, Streamlit, FastAPI, Webex, Zoom
Languages Spoken: English (native), Korean (B2, upper intermediate)